

Several Observations on Particular Properties of the Legendre Polynomials and on the General Orthogonal System

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1 Discrete Transforms via Integral Approximation

Observation: The DFT transforms a set of discrete data points to produce the ‘energy’ contained within each of the integral frequencies via a matrix multiplication. Assembly of these frequencies with such coefficients can approximate the sampled signal.

Goal: Given any complete, orthogonal basis of functions under the integral inner product (Legendre Polynomials, Laguerre Polynomials, Complex Exponentials, etc.) construct a DFT-analog to extract the ‘power’ contained within discrete signal data through a matrix multiplication.

Approach: Given the necessary assumptions of completeness and orthogonality for the basis functions f_j (on $[a,b]$ with weight function w) we have;

$$c_j = \frac{\int_a^b w(t) f_j(t) h(t) dt}{\int_a^b w(t) f_j(t)^2 dt} \rightarrow h(x) \sim \sum_j c_j f_j(x)$$

Let the known denominator integral factor for c_j be $|f_j|^2$. Thus, given a vectorized $\vec{h}(x)$ of discrete values, we need only compute a discrete approximation of the above integral;

$$c_j = \frac{\sum_{x_k}^{[a,b]} w(x_k) f_j(x_k) \psi_k \Delta_k h(x_k)}{|f_j|^2} = \frac{[w] o [F_j] o [\psi \Delta] \cdot h}{|f_j|^2} \quad (1)$$

Where o is a term-by-term multiplication that retains shape, where $[w]_k = w(x_k)$, where $[F_j]_k = F_j(x_k)$, where Δ represents the step-size of the interval between $x_{k+1} - x_k$, and finally where ψ is a vector of weight factors. ψ can be any set of weight parameters requisite for numerical integration: such as $[1, 1, 1, \dots]$ for rectangular approximation $[1, 1/2, 1/2, \dots, 1]$ for trapezoidal, $[1, 2, 4, 2, 4, \dots, 1]/6$ for Simpson’s rule, etc.

Example: Suppose $\vec{h}(x)$ contains $h(x)$ at N equally spaced values on $[-1,1]$, suppose a trapezoidal ψ is taken, and suppose we take f_j to be the j th Legendre Polynomial. Now our ‘Discrete Legendre Transform’ is;

$$\vec{c} = \mathbf{P} \vec{h} ; \mathbf{P}_{lk} = [1, 1/2, \dots, 1]_k(k) \cdot P_l \left(-1 + 2 \frac{k}{N} \right) \cdot \frac{2}{N} \cdot \frac{2l+1}{2}$$

And indeed, constructing the function $h(x) \sim \sum_l P_l(x) c_l = \vec{P}(x) \cdot \mathbf{P} \vec{h}$ forms both an approximation for h and provides the energy of the signal data in the ‘direction’ of the l th Legendre Polynomial - a kind of polynomial analog to the DFT.

Speaking of the DFT, suppose we take the sinusoidal $e^{2\pi i n x}$ on $[0,1]$ with regular spacing of N points under a rectangular approximation. Using the general form of eqn. (1) with these parameters produces the familiar;

$$w_{lk} = e^{2\pi i l \frac{k}{N}} \cdot \frac{1}{N}$$

Excluding the negative n values rescinds completeness of the expansion, but for the sake of comparison.

2 Wronskian Behavior

The Legendre Polynomials arise as eigenfunctions to distinct integer eigenvalues for the Legendre ODE, and consequently do not obey Abel's Identity for the Wronskian between two solutions to the *same* ODE. However, the limiting behavior of consecutive LPolys do exhibit helpful properties;

$$\lim_{n \rightarrow \infty} \mathcal{W}_{2x2} \{P_n(x), P_{n+1}(x)\} = \frac{2}{\pi} \frac{1}{\sqrt{1-x^2}}$$

This is computed by taking the asymptotic;

$$\begin{aligned} P(N, x) &\sim \frac{2 \cos \left(\left(N + \frac{1}{2} \right) \arccos(x) - \frac{\pi}{4} \right)}{\sqrt{2\pi N \sin(\arccos(x))}} \\ &- \frac{2 \cos \left(\left(N + \frac{1}{2} \right) \arccos(x) - \frac{\pi}{4} \right)}{\sqrt{2\pi \sin(\arccos(x))}} \frac{\sqrt{\frac{2}{\pi}} \left(\frac{N + \frac{1}{2} + 1}{N + 1} \right) \sin \left(\frac{\pi}{4} - \left(N + \frac{1}{2} + 1 \right) \arccos(x) \right)}{\sqrt{1-x^2} \sqrt{\sqrt{1-x^2}}} + \dots \\ &\dots + \frac{\sqrt{\frac{2}{\pi}} \left(\frac{N + \frac{1}{2}}{N} \right) \sin \left(\frac{\pi}{4} - \left(N + \frac{1}{2} \right) \arccos(x) \right)}{\sqrt{1-x^2} \sqrt{\sqrt{1-x^2}}} \frac{2 \cos \left(\left(N + \frac{1}{2} + 1 \right) \arccos(x) - \frac{\pi}{4} \right)}{\sqrt{2\pi \sin(\arccos(x))}} \end{aligned}$$

Interestingly, one might expect that this limiting Wronskian would approach the ordinary Wronskian of the differential equation as the components of the differential equations containing N and $N + 1$ would grow asymptotically closer.

$$(1-x^2) P_n''(x) - 2x P_n'(x) + n(n+1) P_n(x) = 0 \rightarrow w_0 e^{\int \frac{-2x}{1-x^2}} = \frac{w_0}{|1-x^2|}$$

But this is not quite the case. More generally, for LPolys spaced by an integer index of q ;

$$\lim_{n \rightarrow \infty} \mathcal{W}_{2x2} \{P_n(x), P_{n+q}(x)\} = \frac{2 \sin(q \arccos(x))}{\pi (1-x^2)}$$

Where a similar evaluation of the asymptotic formula and trig sum-and-difference are employed for simplification.

Perhaps there is compact proof for when and where a limiting Wronskian may exist for a *general* second order linear ODEs, and if the convergent function has a particular connection with the initial equation.

3 Corollaries of Glasser's Theorem

Legendre polynomials, as with many of the standard orthogonal polynomials, have a series of real roots which permit the following family of integral identities; (when convergent)

$$\int_0^\infty f \left(\frac{P(q+1, x)}{P(q, x)} \right) \frac{1}{P(q, x)^2} dx = (q+1) \int_0^\infty f(u) du$$

Leading to the pleasant;

$$\int_0^\infty \frac{1}{P(q, x)^2 + P(q+1, x)^2} dx = \frac{q+1}{2} \pi, \quad q \in \mathbf{N}$$

Beginning with the simpler identity;

$$\int_{-\infty}^\infty f \left(\frac{P(q+1, x)}{P(q, x)} \right) dx = \left(\frac{q+1}{2q+1} \right) \int_{-\infty}^\infty f(x) dx$$

We incite Glasser's Theorem by following polynomial division and partial fractions;

$$\int_{-\infty}^\infty f \left(\frac{2q+1}{q+1} x - \sum_n \frac{a_n}{x - b_n} \right) dx$$

If a_n are all positive, this collapses to the desired form. In terms of residues, this lemma is equivalent to the assertion that the quantity;

$$\left. \frac{\frac{d}{dx} P(q, x)}{P(q+1, x)} \right|_{P(q, x)=0} \leq 0$$

holds for each root of $P(q, x)$. By standard recurrence relations;

$$\left. \frac{\frac{d}{dx} P(n, x)}{P(n+1, x)} \right|_{P(n, x)=0} = \frac{\frac{xP(n, x) - P(n-1, x)}{\frac{x^2-1}{n}}}{\frac{(2n+1)xP(n, x) - nP(n-1, x)}{n+1}} \bigg|_{P(n, x)=0} = \frac{\frac{-1}{\frac{x^2-1}{n}}}{\frac{-n}{n+1}} \leq 0 \quad x \in (-1, 1)$$

We have more forms to get through;

$$\int_{-\infty}^{\infty} f \left(\left(\frac{P(q+1, x)}{P(q, x)} \right)^{-1} \right) \frac{1}{P(q+1, x)^2} dx$$

$$A = \int_{-\infty}^{\infty} f \left(\frac{P(q+1, x)}{P(q, x)} \right) \frac{1}{P(q, x)^{[0,1,2,3]}} dx$$

We also make that, for a similar reason, the Chebshev polynomials also follow this rational-polynomial principal;

$$\int_0^{\infty} f(x) dx = 2 \int_0^{\infty} f \left(\frac{T(q+1, x)}{T(q, x)} \right) dx = \int_0^{\infty} f \left(\frac{T(q+1, x)}{T(q, x)} \right) \frac{1}{T(q, x)^2} dx$$

4 Quadratic Cross Product

It is amusing to note that for the angle of separation between two vectors, u, v , of $\theta = \arccos \left(\frac{u \cdot v}{|u||v|} \right)$, the ‘angle’ between two elements of a functional space can be given by;

$$\theta_{f,g} = \arccos \left(\frac{\int f(x) g(x) w(x) dx}{\int f(x)^2 w(x) dx \cdot \int g(x)^2 w(x) dx} \right)$$

To obtain the cross product, we seek a function h that is orthogonal to both f and g and has a magnitude of length $\sin \theta_{f,g}$, defined above. Much akin to the existence of the cross product on $\mathbb{R}^3$, polynomial crosses can be uniquely constructed for parabolas; (P, Q must be quadratic and monic)

$$P(x) \times Q(x) = (x^2 + ax + b) \sqrt{\frac{|M|^2}{\int_{-1}^1 (t^2 + at + b)^2 dt}} \operatorname{sgn}(P(1) - Q(1))$$

Where a, b are acquired from an orthogonal system (weight 1 on $[-1, 1]$) to be;

$$a = \frac{(2P(0) + \frac{2}{3}) (\frac{2}{5} + \frac{2}{3}Q(0)) + (2Q(0) + \frac{2}{3}) (-\frac{2}{5} - \frac{2}{3}P(0))}{\frac{2}{3}(P(1) - P(0) - 1)(2Q(0) + \frac{2}{3}) + (\frac{2}{3} - \frac{2}{3}Q(1) + \frac{2}{3}Q(0))(2P(0) + \frac{2}{3})}$$

$$b = \frac{-\frac{2}{5} - \frac{2}{3}(P(0) + a(P(1) - P(0) - 1))}{(2P(0) + \frac{2}{3})}$$

Such that $x^2 + ax + b$ is orthogonal to both P and Q . To control the magnitude of the cross product we simply use;

$$|M|^2 = \left(\int_{-1}^1 P(t)^2 dt \right) \left(\int_{-1}^1 Q(t)^2 dt \right) \sin \left(\arccos \left(\frac{\int_{-1}^1 P(t) Q(t) dt}{\left(\int_{-1}^1 P(t)^2 dt \right) \left(\int_{-1}^1 Q(t)^2 dt \right)} \right) \right)$$

The $\operatorname{sgn}(P(1) - Q(1))$ term acts to imbue a simple anti-symmetry under swapping P and Q . Completing some, but not all, of the necessary axioms fundamental to the cross product. It should be noted that none of the triple product or Jacobi identities hold for this purely curiosity of a construction.

5 Cross Products as Legendre Polynomials

The integral inner product of two functions is well established, so what might the cross product take the form of? Heuristically, we must have antisymmetry, orthogonality, and a prescribed magnitude; each of which will be addressed.

Begin by inspection. We desire the cross product of a constant function C and a linear function L .

$$C(x) \times L(x) = s \cdot Q(x) \cdot \text{sgn}$$

Suppose f is the first function to be crossed. The following operation produces -1 or 1 , depending on if f is linear or constant. Thus achieving the antisymmetry constraint. (Take any small $\epsilon > 0$)

$$\text{sgn} = 2 \frac{2f(1) - \int_{-1}^1 f(x) dx}{f(2) - f(0) + \epsilon} - 1$$

Further,

$$Q(x) = \sqrt{\frac{45}{8}} \left(x^2 - \frac{1}{3} \right)$$

Is orthogonal to all C and L with unit norm, $Q \cdot Q = 1$. Finally the scaling factor s must be chosen such that;

$$|C \times L| = s^2 = |C| |L| \sin \left(\arccos \left(\frac{C \cdot L}{|C| |L|} \right) \right)$$

Where ‘the angle between the functions’, is obtained from the arccosine of their dot product. Ultimately, for functions f and g , one of which is linear and one of which is constant, their cross product could then be considered as;

$$\sqrt{\left(\int_{-1}^1 f(t)^2 dt \right) \left(\int_{-1}^1 g(t)^2 dt \right)} \sin \left(\arccos \left(\frac{\int_{-1}^1 f(t) g(t) dt}{\left(\int_{-1}^1 g(t)^2 dt \right) \left(\int_{-1}^1 f(t)^2 dt \right)} \right) \right) \cdot \left(\sqrt{\frac{45}{8}} \left(x^2 - \frac{1}{3} \right) \right) \cdot \left(2 \frac{2f(1) - \int_{-1}^1 f(t) dt}{f(2) - f(0) + 10^{-10}} - 1 \right)$$

More generally of course, the cross product of two polynomials will take the form of a Legendre Polynomial, Q , orthogonal to all polynomials with lesser degree, scaled by the necessary universal s factor derived above.

5.1 Elementary Polynomial Degree Detectors

That there exists such elementary kernels to detect the degree of a polynomial, sgn , in the above arguments, is quite interesting. Two further more trivial examples are;

$$\lim_{x \rightarrow \infty} \frac{\ln(p(x))}{\ln(x)} = \deg p ; \quad \lim_{R \rightarrow \infty} \oint_{|z|=R} \frac{p'(z)}{p(z)} dz = \deg p$$

Moreover, the exactness of the Mittag-Leffler expansion of a rational function occurs only when the function is bounded; ie numerators with polynomials less than or equal to the denominator result in an exact decomposition, whereas all those greater will not be equal.

$$\frac{p(z)}{q(z)} = \frac{p(0)}{q(0)} + \sum_{z_n \in \{q(z)=0\}} \text{Res} \left\{ \frac{p(z)}{q(z)}, z_n \right\} \left(\frac{1}{z - z_n} + \frac{1}{z_n} \right) ; \quad \deg(p) \leq \deg(q)$$

The exactness of Newton-Cotes formulas, such as Simpson’s, Trapezoidal, etc. only hold on polynomials of lesser degree;

$$\int p(x) dx - T_n \{p(x)\} = 0 ; \quad \deg(p) \leq n$$

And the elephant in the room;

$$\int_{-1}^1 P_n(x) p(x) dx = 0 ; \quad \deg(p) < n$$

6 Piecewise Mimicry

That the Legendre Polynomial, when integrated with any lesser-degree polynomial, returns 0 is perhaps one of their most important characteristics. However, it is intriguing to note that one can achieve the same set of constraints (continuity and annihilation) by a set of n piecewise linear functions;

$$\begin{aligned}
 Y_1(x) &= \{-1 \leq x \leq 1 : x\} \\
 Y_2(x) &= \{-1 \leq x \leq 0 : 2x + 1, 0 < x < 1 : -2x + 1\} \\
 Y_3(x) &= \left\{ -1 \leq x \leq -\frac{1}{3} : \frac{13}{4}x + \frac{9}{4}, -\frac{1}{3} \leq x \leq \frac{1}{3} : \frac{-7}{2}x, \frac{1}{3} \leq x \leq 1 : \frac{13}{4}x - \frac{9}{4} \right\} \\
 Y_4(x) &= \left\{ -1 \leq x \leq -\frac{1}{2} : \frac{14}{3}x + \frac{11}{3}, -\frac{1}{2} \leq x \leq 0 : -6x - \frac{5}{3}, 0 \leq x \leq \frac{1}{2} : 6x - \frac{5}{3}, \frac{1}{2} \leq x \leq 1 : -\frac{14}{3}x + \frac{11}{3} \right\} \\
 &\dots
 \end{aligned}$$

When graphed, these expressions even bear a discretized resemblance to the Legendre Polynomials they mimic. If we drop the continuity requirement, we can achieve annihilation of even higher degree.

$$\begin{aligned}
 J_1(x) &= \{-1 \leq x \leq 1 : x\} \\
 J_2(x) &= -2 \{-1 \leq x \leq 0 : -1.5x - 1, 0 \leq x \leq 1 : -1.5x + 1\} \\
 J_3(x) &= -\frac{9}{2} \left\{ -1 \leq x \leq -\frac{1}{3} : -\left(1 + \frac{2}{9}\right)x - 1, -\frac{1}{3} \leq x \leq \frac{1}{3} : -\left(4 + \frac{2}{9}\right)x, \frac{1}{3} \leq x \leq 1 : -\left(1 + \frac{2}{9}\right)x + 1 \right\} \\
 J_4(x) &= -\frac{22}{3} \left\{ -1 \leq x \leq -\frac{1}{2} : -\frac{25}{22}x - 1, -\frac{1}{2} \leq x \leq 0 : -\frac{185}{22}x - \frac{27}{11}, 0 \leq x \leq \frac{1}{2} : -\frac{185}{22}x + \frac{27}{11}, \frac{1}{2} \leq x \leq 1 : -\frac{25}{22}x + 1 \right\}
 \end{aligned}$$

Or for constants, instead of linear functions, on each interval;

$$\begin{aligned}
 C_2(x) &= \{-1 \leq x \leq 0 : -1, 0 \leq x \leq 1 : 1\} \\
 C_3(x) &= \left\{ -1 \leq x \leq -\frac{1}{3} : 1, -\frac{1}{3} \leq x \leq \frac{1}{3} : -2, \frac{1}{3} \leq x \leq 1 : 1 \right\} \\
 C_4(x) &= \left\{ -1 \leq x \leq -\frac{1}{2} : -1, -\frac{1}{2} \leq x \leq 0 : 3, 0 \leq x \leq \frac{1}{2} : -3, \frac{1}{2} \leq x \leq 1 : 1 \right\} \\
 C_8(x) &= \left\{ -1 \leq x \leq -\frac{3}{4} : -1, -\frac{3}{4} \leq x \leq -\frac{1}{2} : 7, -\frac{1}{2} \leq x \leq -\frac{1}{4} : -21, -\frac{1}{4} \leq x \leq 0 : 35, 0 \leq x \leq \frac{1}{4} : -35, \frac{1}{4} \leq x \leq \frac{1}{2} : 21, \frac{1}{2} \leq x \leq \frac{3}{4} : -7, \frac{3}{4} \leq x \leq 1 : 1 \right\}
 \end{aligned}$$

As per usual, that expansions like these are possible represents nothing more than the unremarkable fact that we have many coefficients to wrestle with. Each of these expressions can be found by taking the vector of coefficients $\vec{c}$ from;

$$\vec{c} \in Null \{M\} ; M_{[i, i+q]j} = \int_{-1+\frac{2i}{N}}^{-1+\frac{2i+2}{N}} \left[x^{(q-1)} \dots + x + 1 \right] x^j dx$$

Where $q - 1$ is the degree of the polynomial on each segment. If continuity is to be imposed, the following rows must be appended to M ;

$$M_k = [0, 0 \dots]_k + \left[\left(-1 + \frac{2i}{N} \right)^{(q-1)}, \dots, \left(-1 + \frac{2i}{N} \right), 1 \right] + \left[- \left(-1 + \frac{2i}{N} \right)^{(q-1)}, \dots, - \left(-1 + \frac{2i}{N} \right), -1 \right] + [0, 0 \dots]_{N-k-2}$$

Where each bracketed chunk is of length q (ie 2 for linear polynomials).